Do AI Agents Develop Their Own Social Norms? Language Convergence and Karma Rewards on the Agent-Only Platform Moltbook

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1. Research Question

In late January 2026, Moltbook, a Reddit-style social network populated exclusively by autonomous AI agents, grew within days to hundreds of thousands of registered agents posting, commenting, voting, and accumulating "karma" in thousands of sub-communities ("submolts") [3]. For the first time, machine behavior research can observe a large agent society in the wild rather than in a sandbox. Yet current evidence is contradictory: interaction networks show heavy-tailed activity and attention distributions reminiscent of human platforms [4], while reciprocity is far lower than on Reddit or Bluesky, suggesting agents largely broadcast rather than genuinely interact [3]. Whether agent collectives merely imitate the surface statistics of human social media or develop functional social structure, that is, shared conventions which constrain individual behavior, remains open.

This proposal asks: **To what extent do AI agents on Moltbook develop community-specific linguistic and normative conventions, and does the platform's reputational reward system (karma) act as a selection pressure driving agents toward conformity?** First, norm emergence is a hallmark of human sociality; its appearance in language-model populations under reputational feedback would show that "tribal" convergence is not unique to biological agents but a general property of networked adaptive systems. Second, if karma rewards conformity, agent ecosystems may drift toward homogeneity, the social analogue of model collapse under recursive training on synthetic data [6]. Third, as agentic AI increasingly mediates human information access, self-reinforcing agent norms could accelerate the erosion of a diverse knowledge commons [7].

2. Background and Motivation

Machine behavior calls for studying AI systems as agents with observable behavior, using the tools of the behavioral sciences [1]. Recent work extends this agenda to AI agent behavioral science: agent behavior emerges from the interplay of model internals, environments, social cues, and feedback, and must be studied at the individual, multi-agent, and ecosystem level [2]. First measurement studies of Moltbook map topics and toxicity across 44,411 posts and 12,209 submolts, showing attention concentrating in hubs and polarizing platform-native narratives [3]; network analyses of 39,924 agents and 1.5 million comments document heavy-tailed connectivity [4]. Early evidence even suggests normative dynamics: agents respond to risky instructions posted by other agents with correction, objection, and other informal norm enforcement [5].

What is missing is a test of the underlying mechanism. In human communities, conventions stabilize because conformity is rewarded and deviance is costly; Moltbook's explicit reputational currency (karma) makes this mechanism directly measurable. The stakes are high: if reputational feedback pushes agent populations toward homogeneity, the social dynamics of agent ecosystems connect to model collapse, meaning variance loss under recursive exposure to synthetic outputs [6], and to knowledge collapse, where reliance on convergent AI-generated content degrades society's information ecosystem [7]. Prior work also exposes a confound this proposal addresses by design: security analyses revealed roughly 17,000 human operators behind 1.5 million registered "agents", with no verification that an agent is not a human with a script [8]. A credible study of agent norm formation must therefore pair field data with a controlled, verifiably agent-only replication.

3. Hypotheses

H1 (Linguistic convergence). Within submolts, semantic and stylistic similarity of posts increases over time and exceeds between-submolt similarity, beyond what shared topics explain.

H2 (Reputational selection). Posts closer to their submolt's emergent linguistic norm (embedding centroid) receive more karma, and agents shift subsequent output toward the centroid after positive feedback, creating a conformity feedback loop.

H3 (Norm persistence). In a controlled agent-only environment, conventions that emerge under reputational reward persist after the reward is removed, indicating internalized norms rather than momentary reward-following.

4. Method

A large-scale observational field study (Phase I) is combined with a controlled agent-based experiment (Phase II) that isolates the karma mechanism and eliminates the human-operator confound. H1 and H2 are tested in both phases; H3 only in Phase II.

AI systems and data. Phase I uses (a) the publicly released, annotated dataset of 44,411 Moltbook posts and 12,209 submolts collected before February 2026 [3], and (b) an own longitudinal collection via Moltbook's public API (February to June 2026) covering posts, comments, karma, and timestamps from the 50 most active submolts. Because many accounts are human-operated [8], accounts are filtered on behavioral criteria (near-duplicate template posting, rhythms locked to human waking hours, scripted burst patterns); analyses run with and without filtering as robustness checks. Phase II deploys 30 agents on the open-source OpenClaw framework (10 each backed by a current OpenAI, Anthropic, and open-weight Llama model) in a sandboxed Moltbook-like forum.

Procedure. Phase I is longitudinal over 20 weekly windows. Phase II manipulates one between-condition factor: the karma weight, i.e., how strongly an agent's system prompt instructs it to maximize social approval (none / moderate / high; 3 runs per

condition, 500 interaction rounds). After round 400, the karma signal is switched off in all conditions to test H3.

Variables and measures. Convergence is measured as cosine similarity of sentence embeddings (Sentence-BERT) within vs. between submolts per window, controlling topics via the published taxonomy [3]. Norm proximity is a post's distance to its submolt's rolling embedding centroid; stylistic convergence is additionally captured by type-token ratio and submolt-specific n-gram (slang) adoption. Karma per post is the reputational outcome; covariates include account age, activity, and PageRank centrality in the comment network.

Planned analysis. H1: linear mixed-effects growth models of within- vs. between-submolt similarity (random intercepts for submolts and agents). H2: regression of karma on norm proximity (controlling topic, length, centrality) plus a lagged panel test of whether feedback at t predicts centroid shift at $t+1$; Granger-causality tests probe whether high-centrality "influencer" agents lead community averages. H3: interrupted time-series analysis of convergence before vs. after karma removal, compared across conditions.

5. Expected Results and Contribution

I expect within-submolt similarity to rise significantly faster than between-submolt similarity (H1), karma to reward norm-proximal posts and induce conformity shifts (H2), and sandbox conventions to partially persist after reward removal, with persistence increasing in karma weight (H3). A null result would be equally informative, supporting the view that Moltbook's apparent sociality is surface mimicry without functional norms [3]. The contribution is threefold. Theoretically, it provides one of the first mechanism-level tests of norm emergence in a real agent society, extending machine behavior from individual to collective behavior [1, 2]. For AI safety, it quantifies whether reputational optimization drives agent populations toward homogeneity, linking social feedback loops to model- and knowledge-collapse dynamics [6, 7]. For design, the karma-weight experiment indicates how strongly agent objectives may weight social approval before diversity degrades.

6. Limitations and Ethical Considerations

The main limitation is construct validity in the field data: despite filtering, human-operated accounts cannot be fully excluded [8], and platform instability threatens data continuity, so Phase II carries the causal burden. The sandbox trades ecological validity for control and cannot match Moltbook's scale, though convergence mechanisms should transfer. Embedding similarity is only a proxy for "norms" and is complemented by interpretable measures such as slang adoption. Ethically, only public content is analyzed, agent identifiers are pseudonymized (accounts can be linked to human owners [8]), and no live Moltbook agents are contacted or manipulated. Because conformity mechanisms are dual-use, reporting emphasizes aggregate results, in line with platform terms and TU Berlin's integrity policies.

References

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